MIL-C-52870A(ME)
10 April 1985
SUPERSEDING
MIL-C-52870(ME)
17 July 1975

MILITARY SPECIFICATION

CONVEYOR, FLAT BELT, PORTABLE, GASOLINE-ENGINE DRIVEN

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 30-inch, flat-belt, elevating, portable, gasoline-engine-driven, wheel-mounted, materials-handling conveyor. The conveyor may be used as a single unit or connected in a continuous line.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
VV-G-1690	- Gasoline, Automotive, Low Leaded or Unleaded.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape: Packaging, Waterproof.

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3056 - Gasoline, Automotive, Combat.
MIL-F-3541 - Fittings, Lubrication.
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-G-3859 - Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-P-13983 - Paint, Temporary, Lusterless, Gasoline, Removable.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-B-52761 - Belting, Flat, Conveyor, Rubber Industrial Synthetic Reinforcement.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.

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MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1400	- Engines, Gasoline or Diesel, Methods of Test.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35647	- Padlocks, Key Operated.
MS35829	- Hinge, Continuous (Piano), .075 Thickness, .187 and .250 Pin Diameter.
MS35844	- Cores, Radiators, Engine Cooling Industrial.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

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(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

TIRE & RIM ASSOCIATION, INC. (T&RA)

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association Inc., 3200 West Market Street, Akron, OH 44313.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Z 35.1 - Accident Prevention Signs, Specifications for.
B 20.1 - Safety Standards for Conveyors and Related Equipment.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The portable 60-foot conveyor is continuous gasoline engine-driven belt, supported in a frame to move materials horizontally or up an incline. In the frame, the belt is supported either by idler rollers or steel plates spaced between driving rollers. The conveyor with an elevating mechanism is supported on pneumatic tired wheels and metal pads, and shall handle crated or boxed materials weighing not more than 25 pounds per linear foot of belt at a rate of 90 tons per hour at belt speed of 200 feet per minute up a 8-1/2 degree grade.

3.1.1 Dimension. The conveyor shall conform to the dimensions specified hereunder:

Frame width (max)	3 feet
Conveyor length (min)	60 feet
Conveyor height (max horizontal)	3 feet
Belt width (min)	2-1/2 feet

3.2 First article.

3.2.1 First article model. The contractor shall furnish one or more conveyors as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce conveyors that comply with the requirements of this specification. Examination and tests shall be as specified in section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.5). When specified (see 6.2), the Government will conduct any or all of the first article model examination and tests as specified (see 6.2).

3.3 Safety. The conveyor shall incorporate all safety features and design characteristics of ANSI Standard B20.1. An anti-runway brake or back to stop device shall be provided to prevent personnel injury. Emergency stop devices shall be provided at all operator locations. All rotating, reciprocating, or moving parts which are of such nature or so located as to be hazardous to operating or maintenance personnel shall be fully enclosed or guarded. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed or guarded. Exhaust and other discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. The muffler shall be located no closer than 2 feet from the fuel tank. Where this is not possible fuel shielding will be used to protect the tank from any part of the exhaust system within 1 foot of the tank and insulation and shielding elsewhere. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment.

3.4 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance and repair utilizing common tools and special tools associated with conveyors of this nature. The replacement and

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adjustment of components and accessories shall be accomplished with minimum drainage requirements and with minimum disturbance to other components and accessories of the conveyor. Ease-of-maintenance provisions shall incorporate, to the maximum practicable extent, features insuring operating clearances and facilitating applicable maintenance and servicing operations. Intricate locking devices and controls with threaded fastenings which can be overtorqued by maintenance personnel shall not be used. Covers or plates which must be removed for component adjustment or for component removal shall be equipped with quick-disconnect fastenings. Dimensions of hand-access openings shall be in accordance with MIL-STD-1472 for barehand or light clothing.

3.4.1 Maintenance ratio. The ratio of man-hours of maintenance required to the hours of operation shall not exceed 0.10 hours when tested as specified in 4.6.2.2.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.6 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to the approval of the contracting officer.

3.6.1 Material deterioration and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.6.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.6.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.7 Construction. Construction of the conveyor shall be in accordance with 3.1 and 3.1.1, and as specified herein.

3.7.1 Frame. All framework shall be constructed of formed or rolled structural steel shapes. The top chord angles shall be drilled or punched for the top belt carrier-idler rollers not more than 6 inches on centers. The rollers shall project a minimum of 1/8 inch above the frame. Provisions shall be made for locating the bottom belt-idler rollers at not more than 4 feet on centers, and to support the driving mechanism and take-up adjustment for the driving unit which shall be placed near the center of the conveyor frame. All frame members shall be braced to insure no frame member being stressed beyond 60

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percent of the yield value of the steel when the conveyor is subjected to a uniformly distributed load of 37-1/2 pounds per linear foot. Frame members shall be joined by welding, bolting, or riveting. No portion of the frame shall extend above the level of the belt.

3.7.2 Skirt plate, collapsible. Skirt plate shall be 5 inches high fabricated from No. 12 gage steel minimum mounted on adjustable hinged brackets which will provide necessary clearance between the bottom of the skirt plate and the conveyor belt. The brackets shall permit the lowering of the skirt plates below the line of the belt when conveyor is used for packages wider than the face of the belt. Skirt plates shall be equipped with rubber flashing on bottom or belt side.

3.7.3 Lifting and towing attachments. Permanently affixed lifting eyes shall be installed on frame of conveyor, and shall withstand without damage to any part of the conveyor, 2-1/2 times the resulting vectorial force imposed when conveyor is being lifted. Eye openings and clearance dimensions of the attachment eye shall conform to MIL-STD-209. Lifting attachments may be used for towing attachments.

3.7.4 Connectors. Connectors shall be provided on the head (receiving) and the tail (discharge) ends of the frame, to ensure that the head end is distinguishable from the tail end, so that when assembled with other conveyors all conveyor belts will move in the same direction. Connectors shall be accessible and shall be provided with a locking arrangement to allow for slight variation either vertically or horizontally when connected to another conveyor.

3.7.5 Casters and wheel assemblies. The conveyor shall be equipped with four pneumatic tired industrial wheels to provide mobility. The conveyor shall have two fixed wheel assemblies at the tail end, and two swivel caster wheel assemblies at the head end. The industrial tire and wheel assemblies shall be in accordance with the Tire and Rim Association (T&RA) Yearbook for 5 mph speeds. The wheel assemblies shall be mounted on hubs having a lubricant seal, inner and outer tapered roller bearings, dust cap, and wheel bolts and nuts. Brakes, chocks, or metal pads shall be furnished to hold conveyor in position while in operation.

3.7.6 Elevation mechanism. Provision shall be made to raise and lower the conveyor from horizontal to a minimum of 18 degrees. A positive mechanical lock shall be provided to prevent the conveyor frame from lowering from any set angle.

3.8 Pulleys and idlers.

3.8.1 Pulleys. Pulleys shall be fabricated of pressed steel, electrically welded and mounted on solid shafts.

3.8.1.1 Head pulley. Head pulleys shall be not less than 8 inches in diameter and shall have an accurately crowned face. The shaft shall have a diameter of not less than 1-1/4 inches. Bearings shall be self-alining double bronze bushed fitted into bearing blocks which shall contain a large grease reservoir or self-alining antifriction bearings. All bearings shall be provided with grease fittings as specified in 3.18.2.

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3.8.1.2 Tail pulley. Tail pulleys shall be not less than 8 inches in diameter and shall have an accurately crowned face. The shaft shall have a diameter of not less than 1-1/4 inches. Bearings shall be self-alining double bronze bushed fitted into bearing blocks which shall contain a large grease reservoir or self-alining antifriction bearings. All bearings shall be provided with pressure grease fittings as specified in 3.18.2

3.8.2 Carrier idlers. Carrier idlers shall be drawn steel tubing not less than 4 inches o.d. with electric welded steel heads mounted on a machined shaft. Idlers shall be mounted on free-running antifriction bearings. Carrier idlers shall be mounted on the frame with a maximum spacing of 12 inches between centers.

3.8.2.1 Return idlers. Return idlers shall be similar in all respects to the carrier idlers. There shall be not less than three return idlers to the section.

3.8.3 Takeup. Takeup shall be sufficient for conveyors of this length with a takeup of not less than 4 inches. Takeup shall be screw type, low bearing pressure, of such design to assure the adjusting screw being always in tension and shall be protected from dirt, grit and abuse.

3.9 Rollers.

3.9.1 Belt-idler rollers. The belt-idler roller shall be steel with an outside diameter of between 1.9 and 2.5 inches and a length of not less than 30 inches. The rollers shall be mounted on bearings in accordance with 3.11 and carried on shafts of a size so as not to be damaged when tested in accordance with 4.6.2.2. The top and bottom belt-idler rollers shall be interchangeable and spaced as specified in 3.7.1.

3.9.2 End-transfer rollers. The end-transfer rollers shall be metal tubing with an outside diameter of between 1.9 and 2.5 inches and a length of not less than 30 inches. The rollers shall be mounted as specified in 3.9.1. One or more transfer rollers shall be placed at one end of the conveyor in the space between the end of the frame and the belt return rollers at the head or tail.

3.9.3 Belt-guide rollers. The belt-guide rollers shall be metal tubing with an outside diameter of between 1.9 and 2.5 inches and a length of not less than 6 inches and shall be mounted as specified in 3.9.1. At least 4 rollers shall be placed vertically against each side of the return belt and spaced so as to prevent the belt from tracking against the sides of the frame.

3.9.4 Belt return, takeup, and drive rollers. The rollers at the head and tail ends of the conveyor and the takeup rollers shall be interchangeable and shall be of steel with 12-inch ±1/32-inch outside diameter and 28-inch machined crown face. The drive roller shall be of steel, with 12-inch ±1/32-inch outside diameter and not less than 26-inch machined crown face. All rollers shall be mounted on the shafts as specified in 3.9.1. The drive roller shall be keyed on

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its shaft, but the head and tail return rollers and takeup rollers may be clamped on their shafts. The return belt takeup roller shall be provided with an automatic spring-tension device that operates over the range of stretch and shrink of the conveyor belt. The spring-tension device shall have a screw adjustment accessible for servicing, and shall be protected from damage while being towed.

3.10 Shafting. The shafting for all conveyor belt return, takeup, and drive rollers shall be made from cold-finished steel of adequate strength and size so as not to be damaged when tested in accordance with 4.6.2.2.

3.11 Bearings. The bearings for all rotating shafts and rollers shall be prelubricated. The volume of grease that shall be used to prelubricate the bearing shall be equal to 75 percent of the available airspace between seals.

3.11.1 Takeup adjustment. A takeup adjustment shall be provided for each takeup bearing. The takeup adjustment shall consist of a threaded rod of cold-rolled steel not less than 1 inch in diameter by not less than 36-1/2 inches in length, with a nut to fit the takeup box, a helical steel compression spring, and spring guide bushings. The spring shall be made of 3/8-inch-nominal-diameter spring-steel wire, 2-3/4-inch minimum outside diameter by 14 inches minimum free length. The spring shall compress to a working length of not less than 11 inches by a working load of 350 pounds, and shall compress to a working length of not less than 8 inches by a working load of 750 pounds. One guide bushing of the spring and assembly shall be welded to a plate secured to the frame of the conveyor, and the other guide bushing shall compress the spring by means of a nut and locknut on the rod so as to obtain the desired tension in the belt. The special nuts and bushings shall be made of hot-rolled, mild steel.

3.12 Drive. The conveyor drive shall be a steel roller chain with a 1-inch pitch operating over a cut-tooth steel sprocket on the gear-motor output shaft, and a cut-tooth steel sprocket mounted on the pulley drive shaft. Chain and drive sprocket shall be fully enclosed.

3.13 Conveyor belt. A conveyor belt shall be furnished as an integral part of the conveyor. The belt shall be 30 inches wide and shall be laced with metal fasteners having a pin joint to facilitate dismantling. The belt shall conform to MIL-B-52761, type A, grade 420. Each of the four plies shall be given a coat of latex prior to final assembly. All edges of the finished belt fabric shall be rubber coated.

3.14 Power transmission. Power transmission shall be through enclosed 90 degree, bevel-gear units connected by a drive shaft equipped with universal joints or roller chain couplings. The bevel-gear units shall run in oil and shall be equipped with antifriction bearings. The bevel-gear units shall be mounted inside the conveyor frame: One bevel-gear unit shall be near the power unit, and one shall be at the head pulley. The lower bevel-gear unit shall be driven by the power unit through a short multiple V-belt or chain, and the upper

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bevel-gear unit shall drive the head pulley through a short chain or V-belt drive enclosed in a sheet-metal housing. The drive shafts shall be mounted in antifriction, self-alining bearings.

3.15 Engine. The gasoline engine, including all systems, components, accessories and auxiliaries, shall be heavy duty, 4 cycle standard commercial-industrial type, capable of operating on gasoline conforming to VV-G-1690 and MIL-G-3056, and lubricating oil conforming to MIL-L-2104 and MIL-L-46167. The average oil consumption during first article testing shall not exceed 0.0025 pounds per brake horsepower hour. The temperatures of the oil in the oil sump shall not exceed 250° F and the temperature of the coolant in the radiator top tank shall not exceed 210° F, when the engine is operated under any condition specified herein. The engine shall be the same engine that has been used in the manufacturer's commercial conveyor of the same size and capacity for at least 1 year.

3.15.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the conveyor at any capacity and under any operating conditions specified herein, will not require horsepower in excess of the continuous horsepower rating of the engine as previously published by the engine manufacturer, or 85 percent of the engine manufacturer's previously published maximum horsepower rating at the applicable speed, whichever is less.

3.15.1.1 Engine accessories. Engine accessories shall include the following:

a. A muffler or mufflers.
b. A 15-gallon fuel tank.
c. A 12 volt, negative ground, electric starting system including a 12 volt battery charging system (alternator or generator) of not less than 18 amperes capacity, batteries conforming to MS35000, and a weatherproof battery storage box or compartment treated with acid-resistant paint. Battery connectors shall be suitably protected.
d. Manual cranking system.
e. Housing.
f. Instrument panel with battery-generator or alternator ammeter, fuel quantity indicator, hand choke controls, starter control, high temperature control of engine shut-off type, a safety warning light set to actuate when the engine oil pressure drops below 15 psi, a combination tachometer-hourmeter in accordance with SAE J678d, an engine lubricating oil pressure gage and an engine coolant temperature gage.

3.15.1.2 Engine characteristics. The throttle-governor combination shall not permit speeds that will drive the belt at more than 210 feet per minute (fpm) and shall control the engine at all lower speeds for which the engine is rated for continuous operation. The governor shall be set to drive the belt at 190 to 210 fpm.

3.15.2 Clutch and clutch control. The engine shall be equipped with a hand-operated, overcenter, engagement-type friction clutch and shall be capable

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of operation by an operator standing on the ground. The rated capacity shall be not less than 150 percent of the maximum rated engine torque.

3.15.3 Mounting. The engine and batteries shall be mounted over the tailend section of the conveyor frame on a base and shall have a clearance of not less than 14 inches above the belt. The engine shall be mounted so that it is level when the conveyor is elevated at an angle of 16 degrees.

3.15.4 Sound limits. The noise produced by the conveyor shall not exceed MIL-STD-1474, 5.1.1.1, category D [85 dB(A)] at the operator's position or other occasionally occupied positions when tested in accordance with specification, 4.6.2.5. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 db(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.16 Platform. A platform from which the operator may be able to crank the engine or perform routine lubrication and servicing of the engine shall be provided. The platform shall be provided with an expanded metal surface. The platform shall be hinged along one edge to enable the platform either to be raised to a horizontal position or dropped into a vertical position when not in use. Provision shall be made for holding the platform firmly in both horizontal and vertical positions.

3.17 Electromagnetic interference. The electromagnetic interference emission characteristics of the conveyor shall conform to MIL-STD-461.

3.18 Lubrication. All surfaces requiring lubrication shall be provided with a suitable means for lubricating.

3.18.1 Lubricants. The conveyor shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.18.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where oil holes are applicable, they shall have spring-loaded covers. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.18.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.18.4 Enclosure with integral reservoir of lubricant. Enclosures such as gearcases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, fingerholes, or sightholes to determine the level of the lubricant. Each

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enclosure shall be equipped with a magnetic drainplug not less than 1/2-inch pipe size conforming to MS35844 or MS49006 and with a means for filling the enclosure with lubricant. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the conveyor is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground. Accessibility to the drainplug, the filling means, and the lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts, except that panels and plates equipped with hand-operable quick-disconnect fasteners may be removed.

3.18.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The conveyor shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication because military greases may not be compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.18.6 Oil lubrication. Unless otherwise specified herein all components shall be filled to the operating level with grades of military oils designated for use in the temperature range prevailing at the shipping destination. A tag attached in a conspicuous place shall indicate the temperature range and grade of lubricant used.

3.18.7 Servicing and adjustments. The contractor shall service and adjust each conveyor for correct operation including at least the following: Adjustment of conveyor machinery, checking, and adjustment of the brake system and complete lubrication of the conveyor with types and grades of lubricants recommended in 6.4.

3.19 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.19.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.19.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of

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composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with rule 442 - South Coast Air Quality Management District, El Monte, CA.

3.20 Identification marking. The conveyor shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514, type I, style 3, of composition C, grade A, class 1 material. Transportation data plates conforming to MIL-P-514, type III, of composition C, grade A, class 1 material shall also be furnished and shall indicate the location and capacity of lifting attachments.

3.21 Treatment and painting. The portions of the conveyor and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, semigloss, Forest Green.

3.22 Instruction plates. Each conveyor shall be equipped with instruction plates or diagrams, including safety warnings, cautions, or notice plates describing any special or important procedures to be followed in assembling, operating, or servicing the conveyor. Safety, warning, caution and notice plates shall be in accordance with ANSI Z 35.1, except that the size may be reduced to be compatible with the available space. These plates shall conform to MIL-P-514, type III of composition C, grade A, class 1 material. All plates shall be securely attached to the conveyor with machine screws, bolts and nuts, or rivets and shall be furnished and mounted by the contractor.

3.23 Vehicle marking. The marking of each conveyor shall be in accordance with MIL-STD-642. The gross vehicle weight of the conveyor and the weight of each sectionalized component shall be marked on the side in such a manner as to be discernible. Lifting attachments shall be marked "LIFT HERE".

3.24 Toolbox. Each conveyor shall be provided with a weather-tight, toolbox made of sheet steel not less than 0.071 inch thick. The toolbox lid shall be mounted on hinges conforming to MS35829 with a 1.25-inch diameter brass pin. The toolbox lid shall open to at least 90 degrees from the closed position. The lid shall remain in the fully open position either by its own weight or by holding devices. Two tension-type hook fasteners with strikers shall be provided to keep the lid closed. Not less than 3/4-inch clearance shall be provided between the book lever tip and any other part of the toolbox. The fasteners shall be attached to the toolbox by either bolts or rivets. A hasp which will accommodate a lock conforming to MS35647 shall be provided for locking the toolbox. The toolbox shall be fastened to the conveyor in a protected position. A drainhole with rubber plug shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the tools specified (see 3.26) and on-vehicle equipment (OVE).

3.25 Items to be furnished with the conveyor. No shipment of conveyors shall be made unless repair parts, tools, and accessories are included with the shipment or unless approval for shipment of the conveyor without such items has been received from the contracting officer.

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3.26 Workmanship. The conveyor shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation of the conveyor. Metal used in the fabrication of the conveyor shall be free of kinks and sharp bends. Shearing and chipping shall be done neatly and accurately. Flame cutting of steel may be employed. Burned surfaces of flame-cut steel shall be ground or machined to remove all ash, burrs, and rough edges. All bends shall be made with metal dies of fixtures to insure uniformity in size and shape. All threaded parts shall be in accordance with Federal Standard H28 unless otherwise specified herein. Rivets shall fill the holes completely. Rivet heads shall be full, neatly made, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.26.1 Welders and welding.

3.26.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.26.1.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.26.2 Surfaces. All parts, components, and assemblies including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be cleaned and free of sand, dirt, fins, pits, spruces, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled.

3.26.3 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein. Unless prior approval is obtained from the contracting officer, all tapped holes shall be countersunk before tapping.

3.27 Human factors engineering. The conveyor shall conform to accepted Human Factors Engineering Design Criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to paragraphs 4 (General Requirements), 5.4 (Controls), 5.5 (Labeling), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety) of MIL-STD-1472, as applicable.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection comparison (see 4.7).
d. Inspection of packaging (see 4.8).

4.3 First article examination and test. Prior to examination and test of the conveyor, the following operations shall be performed:

a. Conveyor serviced with oils and greases specified in 3.18 for use in the ambient temperature at which the test will be run.
b. Conveyor run-in as prescribed by the contractor.

4.3.1 Engine tests. Prior to examination and tests of the conveyor, the engine which will be installed in the first article or IPT unit, shall be tested in accordance with test method 2000 of MIL-STD-1400, except test method 2400 shall not apply. One engine out of each lot of 50 or less, shall be tested in accordance with test method series 3000, MIL-STD-1400. Each engine furnished, except the engines selected for the first article tests and production control tests (test method series 3000), shall be tested in accordance with test method series 4000, MIL-STD-1400, except that the engine manufacturer standard commercial run-in may be used in lieu of test method series 4000, provided it equals or exceeds the requirements of test method series 4000, MIL-STD-1400. Failure to meet any requirements in the above tests shall constitute failure of this test.

4.3.2 Examination. The conveyor shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.3 Tests. The conveyor shall be tested as specified in 4.6.2.1 through 4.6.2.5. Failure of any test shall be cause for rejection.

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4.4 First article inspection. When specified, one or more first article conveyors will be selected at random by the Government from conveyors being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of a first article conveyor shall not exclude the remaining conveyors from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the first article test, and failure of any test shall be cause for rejection.

4.4.1 Inspection failure. Failure of a first article conveyor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the first article conveyor and shall be cause for refusal by the Government to continue acceptance of production conveyors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on conveyors previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all conveyors accepted prior to the completion of first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each conveyor shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Test. Each conveyor shall be tested as specified in 4.6.2.6. Failure of the test shall be cause for rejection.

4.6 Inspection procedures.

4.6.1 Examination. The conveyor shall be examined for the following defects. Items 110 through 116 shall be examined while the conveyor is operating.

Major

101. Material not as specified.
102. Parts or components missing or do not function.
103. Assembly incomplete or incorrect.
104. Safety devices or guards missing.
105. Poor fit at joints, seams, or edges; misalinement.
106. Ease of maintenance not as specified.
107. Dimensions not as specified.

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108. Lubrication devices not accessible.
109. Fasteners inadequate or improperly installed.
110. Oil or grease seals leak.
111. Bearings or gearboxes overheat.
112. Shaft misalinement.
113. Belt carrier rolls do not turn.
114. Conveyor does not elevate through the full range.
115. Belt does not run straight on the centerline of the conveyor.
116. Conveyor not in accordance with Federal and ICC Regulations and Standards.
117. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
118. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
119. Contractor does not have documentation available for identification of material, material finishes or treatments.
120. Interchangeability not as specified.
121. Frame not as specified.
122. Skirt plate not as specified.
123. Lifting and towing attachments not as specified.
124. Connectors not as specified.
125. Pulleys and idlers not as specified.
126. Head and tail pulleys not as specified.
127. Carrier and return idlers and takeup not as specified.
128. Lubricants not as specified.
129. Lubrication fittings not as specified.
130. Conveyor belt not as specified.
131. Fungus and moisture resistance not as specified.
132. Toolbox not as specified.
133. Items to be furnished with the conveyor not as specified.
134. Welders and welding not as specified.
135. Welding certification not as specified.
136. Human factors engineering not as specified.

Minor

201. Workmanship not as specified.
202. Treatment and painting not as specified.
203. Identification or other marking, operating and safety instructions, or manuals missing.

4.6.2 Tests.

4.6.2.1 Test conditions (operating). Prior to test, the conveyor shall be lubricated with military oils and greases specified in 3.18. Oils shall be those designated for use in the ambient temperature at which the test is being performed. When services and adjustments are being performed, the conveyor shall be checked for ease of maintenance. Nonconformance to 3.4 shall constitute failure of these tests.

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4.6.2.2 Performance. The conveyor shall be operated for 25 hours at an angle of 11 degrees, 25 hours at an angle of 18 degrees, and 25 hours at any angle in the range of 11 degrees to 18 degrees, at maximum possible loading of the material specified; the tons per hour shall be recorded. The conveyor shall be operated for 25 hours at an angle of 8-1/2 degrees delivering 90 tons per hour of the material specified in 3.1. After each 25 hours of operation the conveyor shall be towed 5 miles over improved roads at speeds not greater than 5 miles per hour. Nonconformance to 3.1, 3.4.1, 3.7.5, 3.7.6, 3.9.1, 3.10, 3.15 or permanent deformation of components shall constitute failure of this test.

4.6.2.3 Engine power. During conduct of the conveyor performance tests specified in 4.6.2.2, the engine rpm and intake manifold vacuum shall be recorded at not less than 30-minute intervals. These readings shall be compared with the engine speed (rpm), brake horsepower and intake manifold vacuum curve, for the continuous brake horsepower condition as established during conduct of test method series 2000, MIL-STD-1400. Any intake manifold vacuum reading recorded during the conveyor performance tests that is lower than the reading listed on the continuous horsepower curve established during conduct of test method 2300, MIL-STD-1400, at the corresponding speed (rpm), shall constitute failure of this test.

4.6.2.4 Electromagnetic interference. The first article model, equipped for the reduction of the electromagnetic interference in accordance with 3.17 shall be tested in accordance with MIL-STD-461. The contractor shall furnish the contracting officer the report of test required by MIL-STD-461 at the time interval specified. Upon approval of the report by the contracting officer and provided all other requirements of the specification are met, the first article model shall be used as a model for all production conveyors. Disapproval of the report shall constitute failure of this test.

4.6.2.5 Sound level measurement. Measure the sound level at the operator's station in accordance with MIL-STD-1474 (steady-state noise) while operating the conveyor at governed speed of 200 feet per minute with the conveyor delivering 90 tons per hour up an 8-1/2 degree grade. Nonconformance to 3.15.4 shall constitute failure of this test.

4.6.2.6 Production test. Each conveyor shall be run for 1 hour at maximum elevation without load and towed for a distance of not less than 5 miles over improved roads at speeds not greater than 5 miles per hour. Evidence of malfunction, permanent deformation, operational difficulties, or breakage shall constitute failure of this test.

4.7 Inspection comparison. The Government may select conveyors at any time during the contract production period and subject these conveyors to the examination and the tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on conveyors selected at random from those which have been accepted by the Government and will not include the previously inspected first article model conveyors. In addition to any test specified as part of the inspection comparison, the Government reserves

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the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison conveyor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison conveyor(s) and shall be cause for refusal by the Government to continue acceptance of production conveyors until evidence has bveen provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on conveyors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all conveyors accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery.

4.8.1 First article pack inspection.

4.8.1.1 Examination. Examine the first article pack for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Test. The first article pack of conveyors for level A shall be subjected to the railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the first article pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the first article pack and shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

117. Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.

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118. Disassembly not as specified for level A or B.
119. Matchmarking not as specified for level A or B.
120. Cloth matchmarking tags not waterproofed as specified for level A.
121. Unprotected surfaces not coated with preservative as specified for level A or B.
122. Engine not preserved in accordance with the referenced document.
123. Belt fasteners not coated with varnish as specified for level A.
124. Conveyor belt not rolled and roll secured as specified for level A or B.
125. Drivebelts not removed or released from tension for level A or B.
126. Gears not preserved as specified for level A or B.
127. Drive chains not preserved as specified for level A or B.
128. Repair parts not preserved in accordance with the referenced document for level A or B.
129. Maintenance tools not preserved in accordance with the referenced document for level A or B.
130. Consolidated packaging not as specified for level A or B.
131. Blocking, bracing, anchoring, and shrouding not in accordance with the referenced documents for level A.
132. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packaging, packing and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4, and shall be subject to surveillance and approval by the Government (see 6.5). The first article pack may be accomplished using either the first article conveyor or a production conveyor. If the first article model conveyor is utilized, any preservation, packaging, and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the first article model conveyor and the production conveyor.

5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. The conveyor shall be prepared for shipment unassembled. Disassembly shall be the minimum necessary to accomplish the packaging specified herein and to safeguard parts known to be subject to pilferage, damage, or loss and to those parts or assemblies that increase cubage and can be removed and reassembled without special tools, in accordance with the contractor's recommendations. Bolts, nuts, pins, and washers involved shall be placed in one of the mating parts and secured to prevent loss. Removed keys shall be secured in the keyway of the pulley or shaft for which intended.

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5.2.1.2 Matchmarking. To facilitate reassembly, components and parts removed from the conveyor shall be matchmarked. Large parts shall be matchmarked by stenciled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-121 or with paint conforming to MIL-P-13983. Small parts shall have matchmarking information on cloth tags conforming to UU-T-81, or on metal tags attached to mating parts. Cloth tags shall be waterproofed in accordance with MIL-STD-129. The tags shall be securely attached to the parts in such a manner as to prevent damage to the preservation or to the parts.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces. Unpainted exterior metal surfaces of component parts requiring the applicabion of a contact preservative in accordance with MIL-P-116, such as bolts, nuts, pins, washers, shafting, rollers and any other unprotected metal surfaces not specified herein shall be coated with type P-1 preservative.

5.2.1.5 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, level A, as specified for type II classification.

5.2.1.6 Conveyor belt. Unprotected belt fasteners shall be coated with varnish conforming to TT-V-121. The belt shall be compactly rolled and the roll shall be secured with two straps conforming to QQ-S-781, type I or IV, light duty, class B, size as applicable. The strapping shall be placed circumferentially on the roll and the straps tensioned so as not to damage the belt.

5.2.1.7 Drivebelts and pulleys. Belts shall be removed or released from tension. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.8 Drive shafts and universal joints. Unpainted exposed metal surfaces of drive shafts and universal joints shall be coated with type P-1 preservative.

5.2.1.9 Gears.

5.2.1.9.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.9.2 Enclosed gears. Gear housings shall be filled to the operating level with the operational lubricant and the gears actuated to assure coating of all interior parts and surfaces with the lubricant.

5.2.1.10 Drive chains.

5.2.1.10.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative

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shall be drained and the chain recoated with type P-1 preservative. Removed chains shall be coiled, preserved as specified above, and the coil secured with soft annealed wire and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to PPP-T-60, type IV. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 preservative.

5.2.1.10.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.2.1.9.2 for enclosed gears.

5.2.1.11 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts and the repair parts placed in the toolbox if space permits.

5.2.1.12 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40, level A and placed in the toolbox if space permits.

5.2.1.13 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 and placed in the toolbox.

5.2.1.14 Consolidated packaging. Packaged and unpackaged small components shall be placed in the toolbox, as space permits, and the toolbox lid closed and secured. Components that will not fit in the toolbox and require the protection of a wood box shall be consolidated packaged together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional. Blocking, bracing and anchoring of the components within the box to prevent movement and damage shall be provided.

5.2.2 Level B. Each complete conveyor shall be preserved and packaged as specified for level A with the following exceptions.

5.2.2.1 Matchmarking. Cloth matchmarking tags shall not require waterproofing.

5.2.2.2 Conveyor belt fasteners. Belt fasteners are not required to be coated with varnish.

5.2.2.3 Drivebelts. Drivebelts shall be removed or released from tension. Removed belts shall be secured in a protected location near the pulley for which intended.

5.2.2.4 Exposed chains. Exposed drive chains shall be preserved as specified in 5.2.1.10.1, except only type P-3 or P-9 preservative shall be applied.

5.2.2.5 Consolidated packaging. Consolidated packaging shall be as specified in 5.2.1.14, except boxes shall be domestic type or class as applicable.

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5.2.3 Level C.

5.2.3.1 Engine. The engine shall be preserved in accordance with the level C, alternate procedure requirements of MIL-E-10062.

5.2.3.2 Repair parts, maintenance tools, technical publications. Repair parts and technical publications shall be preserved and packaged as specified for levels A and B. Maintenance tools shall not require preservation and packaging but shall be placed in the toolbox.

5.2.3.3 Toolbox. The toolbox lid shall be closed and secured.

5.2.3.4 Other components. Preservation and packaging of other components of the conveyor not specifically mentioned in 5.2.3.1, 5.2.3.2, or 5.2.3.3, shall be preserved and packaged to afford protection against deterioration and damage from the contractor to the initial destination. The contractor may use his standard commercial practice when it fulfills these requirements.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. Each complete conveyor shall be packed in a crate conforming to MIL-C-3774, type I, style A. If required, the length requirements shall be waived. The contents shall be blocked, braced, anchored, and protected with a full shroud in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186.

5.3.2 Level C. Each complete conveyor with repair parts, tools, and technical publications shall be packed in a manner to insure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The gasoline power-belt conveyor is intended for use in loading and unloading cartons, boxes, or other small containers in trucks or freight cars, in and around warehouses and docks on hard surface areas.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of first article model and number of conveyors required (see 3.2.1).
c. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2.1).

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d. Time frame required for submission of the preproduction pack (see 5.1).
e. Level of preservation and packaging and level of packing required (see 5.2 and 5.3). Level B preservation, packaging, and packing is intended to provide limited but economical protection and should be specified only when it is determined the conveyors will be held in covered storage.

6.3 First article model. Any changes or deviations of production conveyors from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article model will not relieve the supplier of his obligation to furnish conveyors conforming to this specification.

6.3.1 Noise hazards. If the noise level of the conveyor exceeds 85 dB(A), and the procedures of 3.15.4 have been documented to the satisfaction of the procuring activity, the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 shall apply and include the requirement for hearing protection, the noise level of the conveyor, and the distance from the conveyor at which an 85 dB(A) level will always be met.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the conveyor covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the conveyors in accordance with this specification.

Custodian:
Army - ME

User activity:
Army - CE

Review activity:
DLA - CS

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52870A(ME)	Conveyor, Flat Belt, Portable, Gasoline-Engine-Driven

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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